a lot of other technologies in India are struck at. India still lacks in terms of its hardware manufacturing capabilities. Most of the parts including sensors and motors come from Taiwan, China, and other countries.

We need to understand that when Giga Technologies are produced in bulk, they cost only a fraction of what we pay to source these from other countries. Now, because we are mostly dependent on imports for most of the robotic parts, we pay higher because of import duties and a lot of other such things. The problem becomes greater when the parts you ordered start to malfunction, and you do not have a vendor in your own country.

> WE NEED TO UNDERSTAND THAT WHEN GIGA TECHNOLOGIES ARE PRODUCED IN BULK, THEY COST ONLY A FRACTION OF WHAT WE PAY TO SOURCE THESE FROM OTHER COUNTRIES.

I think the government has done quite a lot and it is now the responsibility of entrepreneurs and industries to do their part as well. A lot of initial steps have been taken, but if you are aiming to make India the hub of autonomous driving hardware manufacturing including the production of lidars, motors, and sensors at established factory lines, then we will have to start by looking at the number of ICE vehicle makers in India, rope them in and their component suppliers as well.

It is a fabulous opportunity; the ministry has said that, the industry has said that, and even the academia has said that too.

**Are the perks of setting up such manufacturing big enough to formulate a plan?**

First, to make such hardware in India, we will have to take aggressive leaps, and the perks of the same go beyond just this industry. We do not talk much about the drone industry, but we should know that the same is very strongly associated with autonomous driving tech. We barely discuss that despite knowing that India needs drones for all its sectors.

Then we have the robotics automation industry, which also resonates with autonomous driving tech. A lot of Indian companies have started using robots imported from the likes of Germany. Once you develop a hardware manufacturing ecosystem you are benefitting the drones, the robotics, and the autonomous driving tech industries. In the larger picture you are creating a growth environment for Industry 4.0 as well.

**How, according to you, can these dots be connected to cash-in on the opportunity?**

The government is already doing a lot on this front. It has set up special economic zones to attract big brands from the world. In fact, it has been done by India at an exceptionally large level already. I think the next important step is to start training people in these fields with special training programs. What this will do is have talent with super-specialised degrees in India. The same will also appeal to foreign companies to set up shop in the country.

Then, of course, there need to be laws introduced that make licensing easy and quick. We should explore ways which can reduce the time between ideation and establishment. There should be a one window solution. But again, a lot of these things have been done already and I seriously have no idea where the industry is getting stuck now!

**You have worked on autonomous driving tech in India and in Europe, how far is the adoption in India as per you?**

When I was working around this tech from 2010 to 2013, we based all our work on the infrastructure that was there in Europe to facilitate autonomous driving. We need to develop that kind of infrastructure to be able to enable adoption of that tech here in India. You cannot even talk about making autonomous tech a reality when you are struggling to make better and safer roads.

These mixed with absence of civilised traffic are the problems that need to be solved immediately. There are people and sectors who will not hesitate in investing into the sector and buy vehicles with such capabilities, but we need to make sure that this tech is safe for India. And as I said earlier, there are use cases in India for which no research is being done anywhere else in the world.

It is also good to admit that L5 is still a distant dream in Europe and USA. L5 is like a binary problem, and even if you solve it 99.9 per cent, you are still left with the problem. The aim in India should be to go after the L3 autonomy. The aim should be to be able to press a button while on a highway and let the tech do all the safe driving for you.

In terms of L3, unfortunately, there is extraordinarily little research being done in India. No vehicle manufacturer is seeing this as a focal point as of now. You will be amazed to know how many modules and other stuff for self-driving tech, to be used in Europe, are being developed in India. Unless the same level of R&D and testing takes place in India, achieving L3 levels will remain a distant dream. This research and testing are a big opportunity for Indian firms.

**Talking about opportunities, is there room for startups in the autonomous driving tech arena?**

There is a big difference between the startup ecosystems of India, Europe, the USA, and the UK. India talks a lot about startups and there are a lot of startups getting registered in India under the Startup India program. The startups are the ones that really dive into the deep technologies space.

However, in countries like the US and the UK you get more access to government and private grants as a startup than you get in India. You also get a lot more investors and help from the academia in those countries as compared to India. This helps startups care more about what they are doing than concentrating on earning bread and butter.

In India, when you are working on a deep tech startup, unfortunately, exceedingly early it goes into the problem of investment. Initial funds given by the government do solve some problems, but they also come with their own set of problems. Unless we can solve the shallow funding problem, we might not see a promising startup in this arena.

Investors need to know that investing in deep-tech companies requires investments for the long run, because when you invest for the long run, the gains are also big! EFY